JianghuSect 2 (JS2-v2.6.0) — Comprehensive Logic, Systems Coherence & Xianxia Contextual Audit Specification (AI Engineering Directive)

- **Target Codebase**: Js2 new (SIM.PATCH_VERSION = '2.6.0', SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting')
- **Intended Reader**: Autonomous AI Software Engineer / Game Systems Architect
- **Audit Scope**: Cross-module static & behavioral trace across sim.js, main.js, ai.js, cognition.js, stats.js, economy.js, cpeco.js, martial.js, proficiency.js, knowledge.js, retraining.js, bottleneck.js, combat.js, cultivation.js, external_market.js, wild_resources.js, content.js, wildlife.js, fishing.js, and visitors_clues.js.

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0. Mandatory Architectural Invariants (Read Before Applying Any Fix)

Any AI agent remediating the defects in this specification **MUST** uphold all five project invariants:

1. **Immutable Human Core Lock**: Never modify js/human_core.js (HC.FINGERPRINT === '155:2f9c9dd7', SHA-256 lock). Any extensions to affinities, skills, or derived attributes must live in wrapper authorities (proficiency.js, martial.js, cultivation.js, stats.js).
2. **40-Script Load Order & Single-File Bundle**: Preserve exact script load parity between index.html and test/load.js, and regenerate JianghuSect_single.html via node tools/build_single.js after any source change.
3. **Deterministic Simulation & Read Purity**: Read-only inspection paths (STATS.explain, UI.*, EXPED.presentChronicleEntry, EXPED.computePeriodDigest) must consume **zero** W.rng calls and must never mutate serialized state (SIM.serialize(W) byte-identical before/after UI inspection).
4. **Closed-Loop Economy Conservation**:
    - **Spirit Stones (stones)** must obey strict Debit === Credit conservation across W.world.resourcePool.stones, W.sect.stones, npc.inv.stones, and visitor.stones. Never mint or destroy stones out of thin air.
    - **External vs. Internal Ledger Separation**: W.sect.ledger (via ECO.earn / ECO.pay) represents **external** Sect cashflow for

`EXPED.computePeriodDigest`. Internal stone transfers between disciples and the Sect must not inflate external income/expense totals.

- **Anti-Minting Provenance**: Player-spawned items (`playerSpawned`, `spawnedMats`, `spawnedPills`, `spawnedTechs`, `spawnedRecipes`, `spawnedCrystals`) must yield `0` CP and `0` Spirit Stones on donation or external sale.

5. **Backward Save Hydration**: Maintain hydration compatibility for saves across `2.4.0` through `2.6.0` (`SIM.SUPPORTED_SAVE_SCHEMAS`). If an existing save contains corrupted cached fields (e.g., inverted `deityPact` modifiers), repair them idempotently during `SIM.deserialize` / `ensure*` hooks.

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1. Executive Defect Triage Matrix

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
CRIT-01	P0 Critical	`js/cultivation.js`	**Sign Inversion on Higher-Realm Deity & Demon Pacts**: `deity.atkMult - 1` turns `+0.24` (`+24%`) into `-0.76` (`-76%` stat cripple) across all 20 Deities & 15 Demon Lords.
CRIT-02	P0 Critical	`js/stats.js`	**Split-Brain Stat Pipeline (`ST.explain` vs `ST.get`)**: `compute()` and `getBaseFast()` diverge on `MECH_PUPPET` (`OVERDRIVE` routed to tank `else` branch; bonuses active at `0%` integrity) and all 6

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			`KARMA_SAMSARA` stances.
CRIT-03	P0 Critical	`js/stats.js`	**Broken Health-Fraction Denominator (`n._hpf`)**: Divides current HP by raw `b.hp` instead of `ST.hpMax(n)`, breaking `<50% HP` berserk/survival passives on geared tanks and penalizing `-hp%` subpaths at 100% full health.
ECO-01	P0 Critical	`js/external_market.js`, `js/economy.js`, `js/content.js`, `js/fishing.js`	**Same-Tick Buy-Then-Liquidate Loop**: `EXT.tradeSect` buys `EQ:` gear without stamping `listedAtDay`, then `EXT.sellStaleSectGear` at the bottom of the same function sees `ageDays = W.day - 0 >= 6` and immediately liquidates the newly bought gear at a ~35% loss. Also affects research prototypes and river container gear.
CTX-01	P1 High	`js/cultivation.js`	**Righteous Sect Masters**

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			Autonomously Devour Rice & Whetstones as "Secret Organs": `monthlyAutonomousTick` falls back to `aliveCults[0]` when specialist paths are absent, eating arbitrary `W.sect.stock` keys (`grain`, `meal`, `ITEM_REPAIR_STONE_*`, breakthrough treasures) for `DEVOURING_FLESH` and spending Sect stones on unaligned shrines without authorization.
CTX-02	P1 High	`js/cultivation.js`	**Unauthorized Sect Treasury Embezzlement for Personal Demon Covenants**: Autonomous disciples debit `160` stones from `W.sect.stones` (instead of personal `n.inv.stones`) to sign personal Blood Covenants with Demon Lords.
CTX-03	P1 High	`js/sim.js`	**Exiled Traitors & Defectors Donate Private Manuals to**

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			Sect Library While Stones & Pets Vanish: `SIM.retire` dumps fleeing exiles' personal unshared manuals/recipes/gear into `W.sect`, while deleting `n.inv.stones` (breaking conservation) and orphaning `n.pets`.
CTX-04	P1 High	`js/sim.js`, `js/economy.js`	**Permanent 30% Stat Rot on Legendary Relics & Remote Workshop Repair**: Non-procedural crafted gear and all 27 Legendary Relics decay to `dur = 1` (`0.30x` stats) and are never auto-repaired (`if (it.procedural)`), while manual `repairGear` permanently destroys `maxDur` down to `20`. Meanwhile, `away` expedition units get remote `mech_workshop` repairs.
AI-01	P1 High	`js/sim.js`, `js/stats.js`, `js/economy.js`,	**Dual Uncoordinated "No-Pill" Vow Systems (`daoOath`**

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
		`js/ai.js`, `js/cpeco.js`, `js/bottleneck.js`	**vs `voluntaryVows`)**: `SIM.hourly` force-feeds pills to injured ascetic disciples, involuntarily shattering their vows; `CPE`/`AI` still buy and auto-consume pills while under oath; `cbBuff` and `btFor` wrappers bypass vow violation checks.
AI-02	P1 High	`js/economy.js`, `js/cultivation.js`	**Broken Multi-Phase Elemental Pills**: `ECO.applyPill` queues into unread `pendingPillPhases` (dead queue); `useConsumableItem` double-applies Phase 1 + top-level stats; `tickMultiPhasePills` passes wrong argument order to `ST.addStatus` (`0.12` as duration -> `1` tick).
AI-03	P1 High	`js/bottleneck.js`	**Personal Breakthrough Pills Bypass Resistance & Mismatch Rules**: `BN.prepare` calls `ECO.applyPill` instead of `ECO.usePill` /

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			`ECO.applyQualityPill`, skipping `BN.pillMult`, `btTaken++` resistance, wrong-path toxicity (`×1.8`), and pill quality `q`.
ECO-02	P1 High	`js/external_market.js`, `js/cpeco.js`	**Sect Confiscates Manuals Bought With Personal Spirit Stones**: `EXT.tradeMember` (`buy-manual`) inserts into both `W.sect.library` (0 CP given) and `n.inv.techs`; `CPE.daily` then strips `n.inv.techs`, forcing the buyer to pay full CP to study their own manual.
ECO-03	P1 High	`js/ai.js`	**Closed-Loop Stone Leaks & External Ledger Pollution**: `A.mine.done` mints `5–18` free stones out of thin air even on `ctx.got === 0`; `AI.kill` destroys 50% of personal stones and logs the remaining 50% via `ECO.earn` as external sect revenue in `W.sect.ledger`.

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
CMB-01	P1 High	`js/combat.js`	**Self-Linked Enemy Tether in `MARIONETTE_SOUL_SWAP`**: Sets `linkedToId: t.id` on target `t`, which `interceptPreDamage` (`linkedToId !== d.id`) ignores, rendering damage mirroring 100% non-functional.
CMB-02	P2 Medium	`js/combat.js`, `js/stats.js`	**Missing Buff ID `'ironSkin'` in `KHMER_YANTRA_WARD`**: `ST.applyNamedBuff(a, 'ironSkin', ...)` silently fails (`null`) because `'ironSkin'` is absent from `ST.BUFF_CATALOG`.
CMB-03	P1 High	`js/visitors_clues.js`	**Copy-Pasted Legendary Armor On-Struck Handler**: `CRANE_CLOUD_AEGIS`, `KIRIN_PHANTOM_COUNTER`, and `XUANWU_UNYIELDING_BASTION` all run `DAGOU_REDIRECTION`'s block—ignoring promised `qiRestorePct` and stun/armorbreak

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			cleanse while dealing unlisted reflect damage labeled `"พลองหยกยืมแรงปัดสวน"`.
CMB-04	P2 Medium	`js/combat.js`	**`COMBAT.leave` Instantly Purges `vowBacklash` Penalty**: `C.leave` strips all `u.sx` entries lacking `fromBuffCatalog:true`, allowing any 1-tick fight/end to erase the 30-minute `vowBacklash` punishment.
AI-04	P2 Medium	`js/ai.js`, `js/cognition.js`, `js/visitors_clues.js`, `js/stats.js`	**Incomplete Integration of 6 Grand Dao Weapon Families & Weapon Conditions**: `ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, `ABACUS` missing from `COG.evaluateAspirations` (`wpnSk`); `AI.pickTeachSkill` lacks `_EXTRA_AFF_MAP`; `simulateGuardianEncounter` hardcodes `'SWORD'` practice; `techT` (+30% ATK/MATK) ignores

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			`conditionMatches`.
AI-05	P2 Medium	`js/wild_resources.js`	**Invalid Office Post IDs in `gatherWildResource`**: Checks `'medicine'`, `'apothecary'`, `'herbs'` instead of canonical `SECT.POSTS` keys (`'herbal'`, `'alchemy'`, `'healer'`).
WLD-01	P1 High	`js/fishing.js`, `js/ai.js`	**Broken Fish Dish Consumption & Dead `studyMult` Buff**: `AI.ACTS.eat.can` ignores fish dishes when `meal === 0`; `eat.start` + `eat.tick` double-feeds `+120` satiety without reserving a meal; `foodBuff.studyMult` is completely unwired; riverbank instant-cooking hardcodes `FOOD_JADE_KOI_SOUP` for incompatible Eel/Dragon Carp.
WLD-02	P2 Medium	`js/fishing.js`	**`SUBPATH_LOOT_TIER` Covers Only 37 of**

ID	Severity	Subsystem / Module	Summary of Logic Bug or Contextual Contradiction
			286 Subpaths: ~249 newer Grand Dao subpaths default to `lootTier = 1` (`Common`), polluting Tier-1 mystery boxes with cosmic/divine subpaths while starving Tiers 2–4.
WLD-03	P2 Medium	`js/visitors_clues.js`	**Missing `CLUE_TEMPLATES` for All 6 Grand Dao Legendary Weapons**: `ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, and `ABACUS` Legendary Weapons advertise `obtainSources: ['CLUE_QUEST', ...]` but have zero clue templates in `CLUE_TEMPLATES`.
WLD-04	P2 Medium	`js/wildlife.js`	**Wild Beasts Wander Onto Impassable Cliffs & Deep Water**: `WILD.tick` 90-tick wander omits `!MAP.blocked(W.map, nx, ny)`, trapping beasts on unpathable tiles and causing `wildHunt` pathfinding abort loops.

2. Detailed Technical Defect Specifications & Remediation Blueprints

[CRIT-01] Sign Inversion on Higher-Realm Deity & Demon Pacts (`-68%` to `-82%` Stat Penalty Instead of Buff)

- **Location**: `js/cultivation.js` — `CULT.HIGHER_REALMS_PANTHEON.DEITIES`, `DEMON_LORDS`, `formDeityPact`, and `getUnitBlessingBonuses`.
- **Root Cause**: In `DEITIES` and `DEMON_LORDS`, stat modifiers are authored as additive fractional deltas (`0.18` to `0.32`), e.g.:
  - `PANGU`: `{ atkMult: 0.24, penBonus: 25 }`
  - `SHIVA_NATARAJA`: `{ matkMult: 0.28, penBonus: 28 }`
  - `VISHNU_PRESERVER`: `{ defMult: 0.24, shieldPct: 0.25 }`
  - `SATAN_WRATH`: `{ atkMult: 0.32, critBonus: 24 }` However, `formDeityPact` and `getUnitBlessingBonuses` evaluate:

```javascript
atkPct: deity.atkMult ? +(deity.atkMult - 1).toFixed(3) : 0,
matkPct: deity.matkMult ? +(deity.matkMult - 1).toFixed(3) : 0,
defPct: deity.defMult ? +(deity.defMult - 1).toFixed(3) : 0,
mdefPct: deity.mdefMult ? +(deity.mdefMult - 1).toFixed(3) : 0,
```

  Because `0.24 - 1 = -0.76`, `stats.js` (`getBaseFast` / `compute`) adds `-0.76` to `p.atk`, **slashing the cultivator's ATK/MATK/DEF/MDEF by `-68%` to `-82%`!**

- **Contextual / Gameplay Impact**: Paying 80 Spirit Stones to form a sacred pact with a supreme deity like Pangu or Zeus cripples an Elder into fighting weaker than a Rank-0 outer initiate.
- **Remediation Blueprint**:
  1. Normalize multiplier conversion via helper `const toPct = v => !v ? 0 : (v >= 1 ? +(v - 1).toFixed(3) : +v.toFixed(3));` in both `formDeityPact` and `getUnitBlessingBonuses`.
  2. Add save-migration repair inside `getUnitBlessingBonuses` / `SIM.deserialize`: if an existing `npc.deityPact` has `atkPct < -0.35` (or `matkPct`/`defPct`/`mdefPct < -0.35`), recompute those fields from `DEITIES[dp.deityId] || DEMON_LORDS[dp.deityId]`.

[CRIT-02] Split-Brain Stat Engine (`ST.explain` UI vs `ST.get` Runtime Combat)

- **Location**: `js/stats.js` (`compute(n, wp)` vs `getBaseFast(n)`) and `js/cultivation.js` (`CULT.SIX_REALM_STANCES`, `CULT.petStatBonuses`).
- **Root Cause**: `stats.js` maintains two separate implementations of character stat aggregation: `compute(n, wp)` (used by `ST.explain` for UI character sheets) and `getBaseFast(n)` (used by `ST.get` in live combat). They have drifted severely:
  1. **`MECH_PUPPET` Construct Mode**:
     - In `compute` (UI): `SIEGE` and `OVERDRIVE` grant `+8% ATK, +8% MATK, +8 PEN`; `SWARM` grants `+10 CAST, +8% ACC (percent), +4 CRIT`; default grants `+10% DEF, +8% MDEF`. Integrity `< 25` applies `-6% DEF`.
     - In `getBaseFast` (Combat): `if (ms.mode === 'SIEGE')` grants `+16% ATK, +16% MATK, +12 PEN, -8% EVA`; `SWARM` grants `+16 CAST, +12 flat ACC, +8% EVA, 0 CRIT`; **`ms.mode === 'OVERDRIVE'` is not handled in the `if/else` chain**, falling into the defensive `else` branch (`+18% DEF, +14% MDEF, +0% ATK/MATK`)! Furthermore, `getBaseFast` applies full mode bonuses even when construct `integrity === 0`.
  2. **`KARMA_SAMSARA` Six-Realm Wheel Stances**:
     - `CULT.SIX_REALM_STANCES` (`cultivation.js:1792`), `compute` (`stats.js:354`), and `getBaseFast` (`stats.js:500`) define **three mutually contradictory stat tables**! For example:
       - `NARAKA`: UI shows `+5 CRIT, +10 PEN`; `SIX_REALM_STANCES` says `+16 PEN, +20 CRIT_DMG`; `getBaseFast` gives `+16 PEN, +12 flat ACC, +10% MATK, 0 CRIT`.
       - `PRETA`: UI shows `+8% EVA, +8 PEN`; `getBaseFast` gives `+12% MATK, +8 flat EVA, 0 PEN`.
       - `ASURA`: UI shows `+8% ATK/MATK, +12 CRIT_DMG`; `getBaseFast` gives `+16% ATK, +14% MATK, +6 CRIT, -6% DEF`.
       - `ANIMAL`/`BEAST`: UI shows `+8% HP, +8 MOVE`; `getBaseFast` gives `+12 MOVE, +12% ATK, +10% EVA, 0% HP` (and if `stance === 'BEAST'` before normalization, `getBaseFast` grants nothing).
- **Remediation Blueprint**:
  1. Extract a single shared function `applyPathStateModifiers(n, emitMod)` used by **both** `compute(n, wp)` and `getBaseFast(n)` so UI tooltips and combat math can never diverge.
  2. Route `ms.mode === 'OVERDRIVE'` alongside `'SIEGE'` (or give `OVERDRIVE` its distinct offensive profile) and gate positive construct mode bonuses on `ms.integrity >= 25`.

3. Read `KARMA_SAMSARA` stance bonuses directly from `CULT.SIX_REALM_STANCES[ks.stance]`.

[CRIT-03] `n._hpf` Health-Fraction Denominator Mismatch Breaks Conditional Passives & Injury Penalties

- **Location**: `js/stats.js` — `hpMaxBase(n)` and `computeFast(n)` (lines 581, 650–653).
- **Root Cause**: `computeFast(n)` computes `n._hpf = clamp(s0.hp / Math.max(1, hpMaxBase(n)), 0, 1)`, where `hpMaxBase(n)` returns `getBaseFast(n).b.hp` (**raw unscaled base HP**, ignoring all percentage `p.hp` and flat `f.hp` bonuses/penalties from armor, relics, meridians, passives, and subpaths).
- **Impact**:
  1. **Geared / Body-Refining Tanks**: If `b.hp = 1000` and gear/meridians/passives raise `ST.hpMax(n)` to `2200`, taking `1100` damage leaves the tank at `1100 / 2200` (**50% HP**), yet `n._hpf` evaluates `1100 / 1000 = 1.1 -> 1.0` (**100% HP**). All `MARTIAL.passiveStats` conditional passives with `hpBelowPct` (e.g., `<40%` or `<50% HP`) fail to trigger until HP drops below `400–500 / 2200` (~18–22% actual HP)!
  2. **Negative-HP Subpaths (`BLOOD_DEMON`, `HEART_SOUL`, etc.)**: When total `p.hp < 0`, `ST.hpMax(n) < b.hp` (e.g., `940 < 1000`). Even at **100% full health (`s0.hp === 940`)**, `n._hpf = 0.94 < 1.0`, permanently inflicting `conditionMods` low-HP stat penalties and disabling `hpAbovePct: 95` passives!
- **Remediation Blueprint**: Compute `n._hpf` against the character's true pre-buff max HP:

```javascript
const base = getBaseFast(n);
const trueMaxHp = Math.max(1, base.hpVal || Math.round(base.b.hp * (1 +
clamp(base.p.hp, -0.8, 10)) + base.f.hp));
n._hpf = clamp(s0.hp / trueMaxHp, 0, 1);
```

[ECO-01] Same-Tick Buy-Then-Liquidate Bug Destroys Newly Purchased Sect Equipment & Research Prototypes

- **Location**: `js/external_market.js` (`EXT.tradeSect`, `EXT.sellStaleSectGear`), `js/economy.js` (`ECO.mintGear`), `js/content.js` (`C.runExperiment`), `js/fishing.js` (`deliverContainerReward`).

- **Root Cause**:
    1. In `EXT.tradeSect(W, v)` (line 319), when the Sect Treasury buys `EQ:<item>` from a visiting merchant for disciples who need an upgrade (`upgraders > 0`), it calls `W.sect.gear.push(ECO.mintGear(it, 1.0))` — omitting `W` and leaving `eq.listedAtDay === undefined`.
    2. At the bottom of the **exact same `EXT.tradeSect(W, v)` call** (line 451), it invokes `EXT.sellStaleSectGear(W, v, { skill })`.
    3. Inside `sellStaleSectGear` (line 472):

```
JavaScript
const listedAt = eq.listedAtDay != null ? eq.listedAtDay : 0;
const ageDays = (W.day || 0) - listedAt;
if (ageDays < graceDays) continue;
```

    Because `W.day` starts at `>= 360` (Year 2+) and `listedAt` defaults to `0`, `ageDays` is `>= 360 >= 6`! And because zero simulation ticks elapsed between line 319 and line 451, no disciple has set `cognition.goal.targetId = eq.uid`. Consequently, **`sellStaleSectGear` immediately liquidates the brand-new equipment back to the same merchant at `0.74x` price and permanently deletes it from the world**, burning ~35% of the Sect's Spirit Stones on the spread while disciples get nothing!

    4. The exact same missing `listedAtDay` bug causes `sellStaleSectGear` to liquidate newly discovered **Experimental Prototype Equipment** (`content.js:415`) and **Sect-opened Mystery Box Equipment** (`fishing.js:689`) on the next trader visit without waiting `graceDays`.
- **Remediation Blueprint**:
    1. Pass `W` to `ECO.mintGear(W, def, q)` across `external_market.js`, `content.js`, `fishing.js`, and `visitors_clues.js`.
    2. Inside `ECO.mintGear`, whenever `W && Number.isFinite(W.day)`, set `if (it.listedAtDay == null) it.listedAtDay = W.day;`.
    3. When pushing any item into `W.sect.gear` (via purchase, craft, donation, or unequip), stamp `it.listedAtDay = W.day`.

[CTX-01] Righteous Sect Masters Autonomously "Devour Secret Organs" From Rice & Whetstones & Spend Sect Stones on Unaligned Shrines

- **Location**: `js/cultivation.js` — `CULT.GRAND_DAO_MECHANICS.monthlyAutonomousTick` (lines 2046–2056, 2085) and `devourResourceForBody` (lines 2270–2282).

- **Root Cause**:
    1. Before iterating over `aliveCults`, `monthlyAutonomousTick` unconditionally executes:

```javascript
const bodyRefiner = aliveCults.find(u => u.cult.path === 'DEVOURING_FLESH' ||
u.cult.path === 'BODY' || u.cult.path === 'DEMON') || aliveCults[0];
this.devourResourceForBody(W, bodyRefiner, null, { free: false });
const graveKeeper = aliveCults.find(u => u.cult.path === 'NETHER_GRAVE' || ...) ||
aliveCults[0];
this.channelGraveyardEcho(W, graveKeeper);
if (... && (W.sect.stones || 0) >= 700) {
  const ritualist = aliveCults.find(...) || aliveCults[0];
  g.SECT.COSMIC_SHRINE.consecrateOffering(W, ritualist, 'HEAVENLY_COURT_DAO',
null, { stoneFallback: 40 });
}
```

    2. Because of `|| aliveCults[0]`, even in a 100% orthodox Sword/Buddhist sect with **zero** `DEVOURING_FLESH`, `NETHER_GRAVE`, or Divine-path cultivators, `aliveCults[0]` (typically the Sect Master) performs eldritch organ devouring, channels necromantic grave echoes, and spends `40` Sect Spirit Stones/month on Cosmic Shrine offerings! And if a `DEVOURING_FLESH` cultivator *is* present, `devourResourceForBody` runs **twice** per month (once before the loop and once inside `for (const n of aliveCults)`).
    3. Worse, `devourResourceForBody` with `itemId === null` selects: `Object.keys(W.sect.stock).find(k => W.sect.stock[k] > 0 && !k.startsWith('MANUAL_PATH_') && !k.startsWith('CRYSTAL_PATH_'))` This indiscriminate predicate eats `'grain'`, `'meal'`, `'ITEM_REPAIR_STONE_4'` (whetstones), `'TRASH_RIVER_DEBRIS'`, or irreplaceable **Breakthrough Treasures (`JD.BT.TREASURES`)** to increment `devouredOrgansCount`!
- **Remediation Blueprint**:
    1. Delete the forced `|| aliveCults[0]` pre-loop block in `monthlyAutonomousTick`; execute Grand Dao path mechanics **only** for living cultivators who actually practice the corresponding `path`.
    2. Restrict `devourResourceForBody` auto-selection strictly to biological beast materials (`MAT_BP_*` or surplus `W.sect.organs` above reserve floors), never food staples, repair stones, pills, or breakthrough treasures.
    3. Require Sect policy/alignment compatibility and `cashReserve` safety before spending `W.sect.stones` on `COSMIC_SHRINE` or `INFERNAL_SHRINE` tributes.

[CTX-02] Low-Rank Disciples Embezzle 160 Sect Treasury Stones to Sign Personal Demon Covenants

- **Location**: `js/cultivation.js` — `monthlyAutonomousTick` (lines 2119–2123) and `CULT.DEMONIC_PANTHEON.formDemonPact` (lines 2500–2514).
- **Root Cause**: In `monthlyAutonomousTick`, any disciple passing `isNpcWillingForDemonPact(n)` (even an outer disciple in an orthodox sect!) triggers `signBloodCovenant` -> `formDemonPact`, which deducts `COVENANT_COST_STONES = 160` directly from **`W.sect.stones`** without checking `n.inv.stones`, `n.rank`, or Sect alignment!
- **Remediation Blueprint**: For autonomous personal pacts (`signBloodCovenant` / `formDemonPact` / `formDeityPact`), require payment from **personal `npc.inv.stones`** unless the character is the Sect Master (`npc.id === W.sect.masterId`) or an authorized Elder in an explicitly Demonic/Pantheon-aligned sect with `W.sect.stones >= cashReserve + cost`.

[CTX-03] Exiled Traitors & Defectors Magically Donate Personal Gear & Private Manuals to the Sect While Stones & Pets Evaporate

- **Location**: `js/sim.js` — `SIM.tick` (line 255) and `SIM.retire(W, n)` (lines 264–285).
- **Root Cause**: When an exiled/defecting NPC finishes walking out the mountain gate (`n.away && n.exiled`), `SIM.retire(W, n)` runs and:
    1. Transfers all `n.equip`, `n.inv.gear`, `n.inv.pills`, `n.inv.mats`, `n.inv.crystals`, `n.inv.subManuals`, and **unshared personal `n.inv.techs` / `n.inv.recipes`** straight into `W.sect` (`W.sect.library` / `W.sect.knownRecipes`)!
    2. Deletes the NPC along with `n.inv.stones` (violating Spirit Stone conservation!) and `n.pets` (orphaning/destroying bonded beasts, unlike `dailyUpdate` which rescues dead owners' pets into `W.sect.beasts`).
- **Contextual Impact**: A traitor fleeing the sect hands over secret manuals the sect never owned at the border, while their money and spirit beasts vanish from existence.
- **Remediation Blueprint**: Pass a `reason` option to `SIM.retire(W, n, { reason })`:
    - If `reason === 'exiled'` or `'defected'`: strip only Sect-issued equipment *at expulsion time* (before walking out); when leaving the map, credit `n.inv.stones` back to `W.world.resourcePool.stones` (preserving closed-loop stone conservation), do **not** auto-grant unlearned personal `n.inv.techs`/`recipes` to `W.sect.library`, and either transfer high-sect-bond `n.pets` to `W.sect.beasts` or release them.

- If `reason === 'ascended'` or `'peaceful_retire'`: bequeath `n.inv.stones` (internal transfer to `W.sect.stones`), gear, manuals, and `n.pets` (`W.sect.beasts`) cleanly to the Sect.

[CTX-04] Permanent 30% Durability Rot on Legendary Relics & Crafted Gear + Remote Expedition Workshop Repairs

- **Location**: `js/sim.js` (`SIM.hourly`, lines 302–335), `js/economy.js` (`ECO.wearGear`, `ECO.repairGear`), `js/stats.js` (line 394).
- **Root Cause**:
  1. `SIM.hourly` wears `it.dur` by `0.5` on **all** equipped gear during combat.
  2. Non-procedural gear (all recipe-forged T1–T7 items and all **27 Unique Legendary Relics**) clamps at `it.dur = 1` in `ECO.wearGear`, where `stats.js` (`clamp(it.dur / 100, 0.3, 1)`) inflicts a **permanent `-70%` stat penalty (`0.30x`)**!
  3. `SIM.hourly` (`if (it.procedural && n.act.id !== 'fight')`) **never repairs non-procedural gear**! And if `ECO.repairGear` is called manually on a Legendary Relic, it permanently reduces `it.maxDur` by `10–25` down to `20`, permanently crippling one-of-a-kind world relics!
  4. Conversely, `SIM.hourly` runs on `!n.dead` **without checking `!n.away`**, so disciples away on distant world expeditions receive hourly `mech_workshop` facility bonuses and Sect-funded gear repairs from hundreds of *li* away.
- **Remediation Blueprint**:
  1. Add `if (n.away) return;` to facility-assisted repairs in `SIM.hourly`.
  2. For `it.legendary || it.unique` items, either make them immune to mundane durability wear (`Spirit-Qi self-tempering`) or allow Forge/Qi maintenance to restore `dur = 100` **without** reducing `maxDur`.
  3. Allow non-procedural crafted equipment (`!it.procedural && !it.legendary`) to be maintained at the Sect Forge (restoring `dur` without permanently bricking `maxDur` when repaired by a skilled Forging disciple, or applying gentle bounded wear only to field-salvaged `procedural` drops).

[AI-01] Two Uncoordinated "No-Pill" Vow Systems (`daoOath` vs `voluntaryVows`) Cause Involuntary Oath-Breaking

- **Location**: `js/cultivation.js` (`OATH_CATALOG.VOW_ABSTAIN_PILLS` / `VEGETARIAN_PACIFIST` vs `VOLUNTARY_VOWS.CATALOG.VOW_NO_MEDICINE`), `js/sim.js` (`SIM.hourly`), `js/economy.js` (`ECO.applyPill`, `ECO.usePill`), `js/stats.js` (`ECO.applyPill` wrapper, `ST.preFight`), `js/bottleneck.js`

(`ECO.usePill` wrapper), `js/cpeco.js` (`pickPurchase`, `autoUse`, `fightUse`, `personalPillUse`), `js/ai.js` (`cultivate`, `breakthrough`), `js/external_market.js` (`memberQuote`).

- **Root Cause**: Two independent systems represent a disciple's oath to abstain from pills, and **neither is consistently respected**:
    1. `SIM.hourly` (`sim.js:347`) force-feeds `P:BASIC_*` pills from `W.sect.stock` to any injured `rank >= 1` NPC (`hp < 50%`) **without checking either vow**, forcibly shattering the disciple's oath (`vowBacklash` + `+28..35 Heart Demon` + `BROKEN_DAO_OATH` injury) while they are passively resting!
    2. `CPE.pickPurchase`, `CPE.autoUse`, `CPE.fightUse`, `A.personalPillUse`, and `EXT.memberQuote` check **neither** vow—so ascetic disciples spend CP/stones buying pills and immediately eat them, breaking their own vows!
    3. `A.cultivate.start`, `A.breakthrough.start`, and `ST.preFight` check `voluntaryVows.VOW_NO_MEDICINE` **only**, ignoring `daoOath` (`VOW_ABSTAIN_PILLS` / `VEGETARIAN_PACIFIST`).
    4. `stats.js:686` (`cbBuff` wrapper on `ECO.applyPill`) and `bottleneck.js:138` (`btFor` wrapper on `ECO.usePill`) return early without calling the underlying `economy.js` functions, **skipping vow-violation checks altogether** on combat pills and sect-stock breakthrough pills!
- **Remediation Blueprint**:
    1. Define `CULT.hasNoPillVow(npc)` returning `true` if `(npc.voluntaryVows?.VOW_NO_MEDICINE?.active && !npc.voluntaryVows.VOW_NO_MEDICINE.violated)` OR `(npc.daoOath?.upheld && (npc.daoOath.id === 'VOW_ABSTAIN_PILLS' || npc.daoOath.id === 'VEGETARIAN_PACIFIST'))`.
    2. Block autonomous pill purchases and routine/hourly pill consumption whenever `CULT.hasNoPillVow(npc)` is true.
    3. Centralize vow-violation enforcement in a single `ECO.enforcePillVowConsequences(W, npc, info)` called by **all** pill consumption paths (`applyPill`, `cbBuff` wrapper, `btFor` wrapper, `useConsumableItem`).

[AI-02] Broken Multi-Phase Elemental Pill Queues, Double Phase-1 Application & Corrupt `ST.addStatus` Call

- **Location**: `js/economy.js` (`ECO.applyPill` line 490, `brewMultiphaseElementalPill` lines 997–1005, `useConsumableItem` lines 1174–1226) and `js/cultivation.js` (`CULT.tickMultiPhasePills` line 1916).
- **Root Cause**:

    1. **Dead Queue**: `ECO.applyPill` queues delayed phases into `npc.ext.pendingPillPhases`, which is **never ticked anywhere** in the codebase (`tickMultiPhasePills` only reads `npc.ext.multiPhasePills`).
    2. **Double Phase-1 Effect**: `brewMultiphaseElementalPill` defines *both* top-level `healPct`/`qiPct`/`cultExp`/`grantBuff` AND `phases[0]`. `ECO.useConsumableItem` applies the top-level fields **and then immediately applies `phases[0]`**, doubling the instant heal/Qi/EXP and overwriting `phases[0].grantBuff`!
    3. **Wrong `ST.addStatus` Signature on Phase 2**: `cultivation.js:1916` calls: `g.STATS.addStatus(npc, p2.aftermathStatus, p2.aftermathVal || 0.12, p2.aftermathDur || 4, tNow, mp.itemId)` Whereas `ST.addStatus` (`stats.js:285`) has signature `(u, k, dur, extra, now)`. Passing `0.12` as `dur` clamps duration to **1 tick** instead of 4+ ticks, and passing number `4` as `extra` discards metadata!
- **Remediation Blueprint**:
    1. Route `ECO.applyPill`'s `e.phases` scheduling into `npc.ext.multiPhasePills` (using the exact schema expected by `CULT.tickMultiPhasePills`) and migrate any legacy `pendingPillPhases` entries.
    2. In `ECO.useConsumableItem`, when `Array.isArray(def.phases) && def.phases.length >= 2`, execute **only** `phases[0]` on consumption and queue `phases[1]` (do not also apply duplicate top-level summary fields).
    3. Fix line 1916 to `g.STATS.addStatus(npc, p2.aftermathStatus, p2.aftermathDur || 12, { val: p2.aftermathVal || 0.12, src: mp.itemId || 'MULTI_PHASE_PILL', persistAfterFight: true }, tNow)`.

[AI-03] Personal Breakthrough Pills (`P:BT_*`) Bypass Resistance (`btTaken`), Mismatch Toxicity & Quality

- **Location**: `js/bottleneck.js` (`BN.prepare` line 88, `ECO.usePill` wrapper line 138).
- **Root Cause**: `bottleneck.js` only wraps `ECO.usePill` (direct sect-stock consumption) with `BN.pillMult(c, e.btFor)`, `c.btTaken++`, and `1.8x` mismatch toxicity. However, disciples almost always hold `P:BT_*` in `n.inv.pills` (via `BN.daily` grant or `CPE.shop`), which `BN.prepare` consumes via `ECO.applyPill(W, n, k)`—completely bypassing `BN.pillMult`, drug resistance, mismatch penalties, and `n.inv.pillQuality[k]`!
- **Remediation Blueprint**: Move `btFor` resolution (`BN.pillMult`, `c.btTaken++`, mismatch toxicity `×1.8`) into `ECO.applyPill` (scaled by `ECO.applyQualityPill` when consumed via `BN.prepare`).

[ECO-02] Disciples Buying Manuals From Visiting Merchants Have Their Manuals Confiscated by the Sect

- **Location**: `js/external_market.js` (`EXT.tradeMember` -> `buy-manual`, lines 614–617) and `js/cpeco.js` (`CPE.daily`, line 368).
- **Root Cause**: When a disciple spends personal `n.inv.stones` to buy `TECH_MANUAL:<tid>` from a merchant, `EXT.tradeMember` pushes `tid` into **`W.sect.library`** (giving the Sect a free copy for 0 CP) *and* into `n.inv.techs`. On the next `CPE.daily`, `n.inv.techs` filters out any `id` already in `W.sect.library` (`!S.library.includes(id)`), deleting the disciple's scroll and forcing them to pay full `CPE.techPrice` in Contribution Points to study the manual they bought with their own Spirit Stones!
- **Remediation Blueprint**:
  1. In `EXT.tradeMember` (`buy-manual`), add `tid` **only** to `n.inv.techs` (never directly to `W.sect.library`).
  2. Allow `A.study` / `MARTIAL.startStudy` to consume or study a manual held in `n.inv.techs` for **0 CP fee**, or let the disciple later donate it to `W.sect.library` in exchange for `CPE.techReward` CP (matching personal recipe scroll lifecycle in `content.js`).

[ECO-03] Unbacked Spirit Stone Minting in `A.mine.done` & Inheritance Destruction / Ledger Pollution in `AI.kill`

- **Location**: `js/ai.js` (`A.mine.done` line 592, `A.shop.done` line 701, `AI.kill` line 856).
- **Root Cause**:
  1. `A.mine.done` grants `Math.min(18, Math.max(5, (ctx.got || 1) * 6))` personal Spirit Stones to any disciple with `!CULT.checkTrackGates(W, n).resourceOk`—even when `ctx.got === 0`, and **without debiting `W.world.resourcePool.stones`**!
  2. `AI.kill` destroys `50%` of a deceased disciple's `n.inv.stones` and passes the other `50%` into `ECO.earn(W, ...)`—which records an entry in `W.sect.ledger`, polluting `EXPED.computePeriodDigest`'s external Spirit Stone income metrics with internal inheritance transfers (and `A.shop.done` similarly calls `ECO.earn` on internal disciple purchases).
- **Remediation Blueprint**:
  1. Require `ctx.got > 0` in `A.mine.done` and transfer stones strictly from `W.world.resourcePool.stones` (clamped to available pool balance).

2. When a disciple dies inside the Sect (`!n.away`), transfer 100% of `n.inv.stones` to `W.sect.stones` via an internal ledger tag (`internal: true`) excluded from external revenue in `EXPED.computePeriodDigest`; if they die unrecovered outside (`n.away`), return unrecovered stones to `W.world.resourcePool.stones`.

[CMB-01] `MARIONETTE_SOUL_SWAP` Tethers Target Enemy to Itself (`linkedToId: t.id`)

- **Location**: `js/combat.js` — `COMPOSABLE_ENGINE.executeAction` (`case 'MARIONETTE_SOUL_SWAP'`, line 1435).
- **Root Cause**: `ST.addStatus(t, 'tetheredEnemy', 16, { linkedToId: t.id, shareRatio: 0.30 }, tNow)` sets `linkedToId` to `t.id` (the target itself). In `interceptPreDamage` (`line 1808`), `if (tether && tether.linkedToId && tether.linkedToId !== d.id)` explicitly rejects self-tethers, so `MARIONETTE_SOUL_SWAP`'s tether never mirrors a single point of damage!
- **Remediation Blueprint**: Select a second living hostile unit `const other = f ? C.enemies(W, f, a.id).find(u => u && u.id !== t.id && C.alive(u)) : null;` and link `t` <-> `other` bidirectionally (as in `preExecuteSkill` line 1696); if solo target, apply a spirit-puppet vulnerability rider instead of a dead self-tether.

[CMB-02] Missing Buff `'ironSkin'` in `KHMER_YANTRA_WARD` (`OCCULT_MAGE`)

- **Location**: `js/combat.js` (line 1538) and `js/stats.js` (`ST.BUFF_CATALOG`, lines 132–186).
- **Root Cause**: `KHMER_YANTRA_WARD` calls `ST.applyNamedBuff(a, 'ironSkin', 22, tNow, { casterId: a.id })`, but `'ironSkin'` does not exist in `ST.BUFF_CATALOG`, so `applyNamedBuff` returns `null`.
- **Remediation Blueprint**: Add `ironSkin: { id: 'ironSkin', nameTh: 'คงกระพันชาตรีอาคมขอม', icon: '🛡️', defaultDur: 22, p: { def: 0.20, mdef: 0.14 }, f: {} }` to `ST.BUFF_CATALOG` in `js/stats.js` (and `SK.STATUS` if rendered as a status badge).

[CMB-03] Legendary Armors Copy-Paste `EFX_LEG_DAGOU_REDIRECTION` Struck Handler

- **Location**: `js/visitors_clues.js` — `VC.runWeaponOnStruckProcs` (lines 721–742).
- **Root Cause**: All three Legendary Armors (`EFX_LEG_CRANE_CLOUD_AEGIS`, `EFX_LEG_KIRIN_PHANTOM_COUNTER`, `EFX_LEG_XUANWU_UNYIELDING_BASTION`) were appended to `EFX_LEG_DAGOU_REDIRECTION`'s `if` condition without per-armor branching:
  - `LEG_ARM_CELESTIAL_CRANE_ROBE` (`params: { shieldHpPct: 0.18, qiRestorePct: 0.05, grantBuff: 'swiftWind' }`) **never restores the promised 5% Qi**, yet deals unlisted 15% reflect damage with hardcoded floating text `'พลองหยกยืมแรงปัดสวน'` (*"Jade Staff Redirects Force"*!), slows the attacker, and grants unlisted `evadeUp`!
  - `LEG_ARM_IMMORTAL_TORTOISE_PLATE` promises cleansing `armorbreak`/`stun` in `descTh`, which is never executed, and also displays `'พลองหยกยืมแรงปัดสวน'`.
- **Remediation Blueprint**: Implement explicit branches per `fx.effectId` in `VC.runWeaponOnStruckProcs`:
  - Honor `p.qiRestorePct` when present.
  - Cleanse `armorbreak` and `stun` on `EFX_LEG_XUANWU_UNYIELDING_BASTION`.
  - Reflect/redirect damage **only** when `(p.redirectRatio || p.reflectRatio) > 0`, using `fx.nameTh` as the combat damage note instead of `'พลองหยกยืมแรงปัดสวน'`.
  - Apply `p.selfStatus` / `p.targetStatus` **only** when defined in `p` (do not default to `'evadeUp'` / `'slow'` for armors that don't specify them).

[CMB-04] `COMBAT.leave` Instantly Purges `vowBacklash` After Any Fight

- **Location**: `js/combat.js` — `C.leave` (line 1032).
- **Root Cause**: `C.leave` filters `n.sx = n.sx.filter(x => x && x.fromBuffCatalog && x.until > W.t);`. When an NPC breaks `VOW_NO_RETREAT` (by fleeing!) or `VOW_ASCETIC_PRECEPT` (by casting a forbidden art in combat!), `violateVow` applies `ST.addStatus(npc, 'vowBacklash', 30, ...)` without `fromBuffCatalog: true`. Seconds later when the fight concludes, `C.leave` **erases `vowBacklash` immediately**!
- **Remediation Blueprint**: Set `persistAfterFight: true` when applying `vowBacklash` (and multi-phase pill aftermath statuses), and update `C.leave` to retain `x && (x.fromBuffCatalog || x.persistAfterFight) && x.until > W.t`.

[AI-04] Incomplete Support for the 6 Grand Dao Weapon Families & Weapon-Gated Technique Stats

- **Location**: `js/cognition.js` (line 100), `js/ai.js` (line 803), `js/visitors_clues.js` (line 2116), `js/stats.js` (lines 303, 442).
- **Root Cause**:
  1. `COG.evaluateAspirations` computes `wpnSk` over only 10 weapons, omitting `ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, `ABACUS`.
  2. `AI.pickTeachSkill` queries `HC.eff(s, 'affinities.' + b)` without mapping extra weapons (`BOW`, `ZITHER`, `ROSARY`, `CLAW`, etc.) to valid locked-`HC` affinity keys, returning `0` and biasing teaching toward the 4 original weapons.
  3. `VC.simulateGuardianEncounter` hardcodes `skillPractice(m, 'SWORD', 1.2, 1.1)` on victory even for spear, fist, bow, or zither wielders.
  4. `stats.js` (`compute` line 303 & `getBaseFast` line 442) adds `techT += t.tier` (up to `+30%` global ATK & MATK) from learned `COMBAT_TECHNIQUE` entries *before* checking `MARTIAL.conditionMatches(n, t)`, granting sword-manual attack bonuses to barehanded disciples.
- **Remediation Blueprint**:
  1. Compute `wpnSk` across `(g.MARTIAL && g.MARTIAL.SKILL_GROUPS ? g.MARTIAL.SKILL_GROUPS.weapon : [...])`.
  2. Use `MC_AFF_MAP` / `_EXTRA_AFF_MAP` inside `AI.pickTeachSkill`.
  3. Practice `({ FIST: 'FIST_PALM' }[M.equippedType(m)] || M.equippedType(m))` in `simulateGuardianEncounter`.
  4. Check `if (g.MARTIAL && !g.MARTIAL.conditionMatches(n, t))` *before* incrementing `techT` in both `compute` and `getBaseFast`.

[AI-05] Wrong Office Post IDs in `AI.ACTS.gatherWildResource.score`

- **Location**: `js/wild_resources.js` (line 319).
- **Root Cause**: Checks `if (p === 'medicine' || p === 'apothecary' || p === 'herbs')`, none of which exist in `SECT.POSTS` (`'healer'`, `'alchemy'`, `'herbal'`).
- **Remediation Blueprint**: Check `if (p === 'herbal' || p === 'alchemy' || p === 'healer' || n.assist === 'herbal' || n.assist === 'alchemy')`.

[WLD-01] Sect Starves Next to Cooked Fish Dishes, Double-Eats Satiety, and `foodBuff.studyMult` Is Dead Code

- **Location**: `js/fishing.js` (`AI.ACTS.eat` hook lines 895–921, `handlePostFishingInventory` lines 972–978, `CULT.rate` hook lines 1137–1147) and `js/ai.js` (`A.eat`, `A.study`).
- **Root Cause**:
  1. `A.eat.can` (`ai.js:304`) checks `ECO.count(W, 'meal') >= 1` and was **not** wrapped by `fishing.js`. When `meal === 0`, disciples cannot trigger `eat` even if `W.sect.stock` or `n.inv.mats` contains cooked `FISH_DISHES`!
  2. When `meal >= 1` and `A.eat.start` eats a fish dish (`+34..45` satiety), it returns `true` without `origStartEat` (`ECO.reserveMeal`), and then `A.eat.tick` still runs for 18 ticks adding `+4.6/tick` (`+82.8` extra free satiety without consuming a `meal`)!
  3. `FOOD_JADE_KOI_SOUP` and `FOOD_DRAGON_CARP_BANQUET` define `buff.studyMult = 1.12`, and line 1136's comment states it hooks both cultivation and study speed, **but `studyMult` is never read anywhere in `ai.js`, `cpeco.js`, or `proficiency.js`**.
  4. `handlePostFishingInventory` (`line 973`) hardcodes `dishId = sp.tier >= 3 ? 'FOOD_JADE_KOI_SOUP' : 'FOOD_STEAMED_SPIRIT_FISH'`, which fails for `MAT_FISH_SPIRIT_EEL_3` (`FOOD_EEL_VITALITY_BROTH`) and Tier-5 fish (`FOOD_DRAGON_CARP_BANQUET`), and instantaneously cooks hotpot banquets at the riverbank without visiting the canteen!
- **Remediation Blueprint**:
  1. Wrap `AI.ACTS.eat.can` to also return `true` when `!FISHING.activeFoodBuff(W, n)` (or `satiety < 85`) and any `FISH_DISHES` exist in `n.inv.mats` or `W.sect.stock`.
  2. When `A.eat.start` consumes a fish dish, mark `ctx.ateFishDish = true` and in `A.eat.tick` skip the `+4.6/tick` free `meal` satiety gain (or return `'done'` after eating).
  3. Wire `const fb = g.FISHING && g.FISHING.activeFoodBuff ? g.FISHING.activeFoodBuff(W, n) : null; const studyMult = (fb && fb.studyMult) || 1;` into `A.study.tick` and `A.guidedStudy.tick`.
  4. Map `sp.recipes.find(rid => FISHING.DISH_BY_ID[rid])` instead of hardcoding `'FOOD_JADE_KOI_SOUP'`, and require canteen proximity (or deposit raw fish for `A.cook`) rather than cooking banquets instantaneously in-pocket.

[WLD-02] `SUBPATH_LOOT_TIER` Hardcodes Only 37 Legacy Subpaths Out of 286

- **Location**: `js/fishing.js` (`SUBPATH_LOOT_TIER` lines 406–420 and `buildOrdinaryPoolForCategoryAndTier` line 597).
- **Root Cause**: `SUBPATH_LOOT_TIER` only lists 37 v1.x subpaths; all ~249 newer Grand Dao subpaths fall back to `SUBPATH_LOOT_TIER[s.id] || 1`, crowding Tier 1 (`56%` drop weight) with high-concept Cosmic/Chaos/Divine subpaths while Tiers 2–4 only contain 22 legacy subpaths.
- **Remediation Blueprint**: Deterministically classify all 286 subpaths into `lootTier 1..4` (using `SUBPATH_LOOT_TIER[s.id]` if present, otherwise deriving `1..4` from subpath index/archetype/path tier so every Grand Dao path distributes evenly across Tiers 1–4).

[WLD-03] Missing `CLUE_TEMPLATES` for All 6 Grand Dao Legendary Weapons

- **Location**: `js/visitors_clues.js` (`LEGENDARY_WEAPONS` lines 356–490, `obtainSources` line 540, `CLUE_TEMPLATES` lines 765–1665, `VC.validateCatalog` line 2566).
- **Root Cause**: All 6 Grand Dao Legendary Weapons (`LEG_WPN_BODHI_NIRVANA_ROSARY`, `LEG_WPN_TAOTIE_DEVOURING_CLAW`, `LEG_WPN_BAIHUA_SOUL_NEEDLE`, `LEG_WPN_BAJIAO_TEMPEST_FAN`, `LEG_WPN_HETU_LUOSHU_ASTROLABE`, `LEG_WPN_CAISHEN_GOLDEN_ABACUS`) are stamped with `obtainSources: ['CLUE_QUEST', 'SECRET_REALM', 'RELIC_INHERITANCE']`, yet zero `CLUE_TEMPLATES` target them.
- **Remediation Blueprint**:
  1. Author 6 multi-step `CLUE_TEMPLATES` (`CLUE_LEG_WPN_BODHI_NIRVANA_ROSARY`, etc.) testing each weapon's matching skill (`ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, `ABACUS`) and thematic region.
  2. Add an invariant check in `VC.validateCatalog()` verifying every item in `LEGENDARY_EQUIPMENT` whose `obtainSources` includes `'CLUE_QUEST'` has at least one `CLUE_TEMPLATES` entry with `targetId === item.id`.

[WLD-04] On-Map Wild Beasts Wander Onto Impassable Cliffs & Water Tiles

- **Location**: `js/wildlife.js` — `WILD.tick` (lines 244–253).

- **Root Cause**: `WILD.spawn` checks `!MAP.blocked(W.map, x, y)`, but the 90-tick wander step only checks `isOutsideWall(W.map, nx, ny)` and `isInSafeBelt`, allowing beasts to step onto `CLIFF`, `WALL`, and `WATER` tiles (`MAP.blocked === true`), breaking `AI.ACTS.wildHunt` pathfinding (`MAP.findPath` returns `null` -> `AI.cancelAct`).
- **Remediation Blueprint**: Update line 247 to:

```javascript
if (isOutsideWall(W.map, nx, ny) && !MAP.blocked(W.map, nx, ny) &&
(!MAP.isReachableFromGate || MAP.isReachableFromGate(W.map, Math.floor(nx),
Math.floor(ny)))) {
```

---

3. Recommended Patch Execution & QA Verification Protocol

1. **Phase 1 — Math & Stat Authority Unification (`CRIT-01`, `CRIT-02`, `CRIT-03`, `CMB-02`)**:
   - Fix `HIGHER_REALMS_PANTHEON` `toPct` conversion + save migration.
   - Unify `stats.js` `compute` and `getBaseFast` path-state helpers and `n._hpf` denominator; add `ironSkin` to `ST.BUFF_CATALOG`.
2. **Phase 2 — Closed-Loop Economy & Equipment Lifecycle (`ECO-01`, `ECO-02`, `ECO-03`, `CTX-03`, `CTX-04`)**:
   - Stamp `listedAtDay` in `ECO.mintGear` and all `W.sect.gear` insertions.
   - Fix personal manual purchase ownership, `A.mine.done` pool debit, `AI.kill` internal stone accounting, `SIM.retire` exile vs. retirement branching, and Legendary Relic durability preservation + `!n.away` workshop gating.
3. **Phase 3 — Autonomous AI, Oaths & Multi-Phase Pills (`CTX-01`, `CTX-02`, `AI-01`, `AI-02`, `AI-03`, `AI-04`, `AI-05`)**:
   - Remove `|| aliveCults[0]` fallbacks in `monthlyAutonomousTick`, restrict `devourResourceForBody` to beast materials, and charge personal stones for personal Demon/Deity pacts.
   - Unify `CULT.hasNoPillVow`, multi-phase pill queueing + `ST.addStatus` signature, `btFor` personal pill scaling, 6 Grand Dao weapon proficiency hooks, and `gatherWildResource` post IDs.
4. **Phase 4 — Combat Procs, Ecology, Fishing & Clue Templates (`CMB-01`, `CMB-03`, `CMB-04`, `WLD-01`, `WLD-02`, `WLD-03`, `WLD-04`)**:
   - Fix `MARIONETTE_SOUL_SWAP` enemy tether, per-armor `runWeaponOnStruckProcs`, `vowBacklash` persistence in `C.leave`, `fishing.js` `eat` & `studyMult` hooks, `SUBPATH_LOOT_TIER` coverage, 6 Grand Dao weapon clue chains, and `WILD.tick` `MAP.blocked` guard.
5. **Phase 5 — Mandatory Verification Suite**:

- Run `node test/core_lock.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
- Run `node tools/validate_data_schema.js` (`Errors: 0, Warnings: 0`).
- Add a dedicated regression test suite `test/v270_logic_and_contextual_audit_fixes_suite.js` asserting all 23 fixes.
- Rebuild single-file bundle via `node tools/build_single.js` and run `node tools/run_parallel_tests.js` (100% pass, 0 timeouts).